

Student marks memo:

```
#include<stdio.h>

#include<conio.h>

void main()

{

char  rollno[20], name[30],
gen[20],dob[15],father[20],mother[20],school[30],id1[30
],id2[30],

subj[6][10]={"Tel","Eng","Hin","Mat","Sci","Soc"} ;

int  sub[6],tot=0, i,p=1;float avg;

clrscr();

printf("Enter rollno "); scanf("%s",rollno);

printf("Enter stu name "); scanf(" %[^\\n]",name);

printf("Enter Gender "); scanf(" %[^\\n]",gen);

printf("Enter Date of birth "); scanf(" %[^\\n]",dob);

printf("Enter Father name "); scanf(" %[^\\n]",father);

printf("Enter Mother name "); scanf(" %[^\\n]",mother);
```

```
printf("Enter School name "); scanf(" %[^\n]",school);

printf("Enter Identification mark 1 "); scanf("
%[^\n]",id1);

printf("Enter Identification mark 2 "); scanf("
%[^\n]",id2);

printf("Enter 6 sub marks ");

for(i=0;i<6;i++)

{

scanf("%d",&sub[i]);tot+=sub[i]; if(sub[i]<35)p=0;

}

avg=tot/6.0;

puts("_____
_____");

puts("\t\t\t BOARD OF SECONDARY EDUCATION");

puts("\t\t\t   ANDHRA PRADESH");

puts("_____
_____");

printf("Rollno: %s\t Name: %s\n",rollno,name);
```

```
printf("Gender: %s\t Date of Birth: %s\n",gen,dob);
printf("Father: %s\t Mother: %s\n",father,mother);
printf("School: %s\n",school);
puts("_____
_____");
puts("Identification Marks");
puts("_____
_____");
printf("1. %s\n",id1);
printf("2. %s\n",id2);
puts("_____
_____");
puts("Subject\tMarks\tPass/Fail");
puts("_____
_____");
for(i=0;i<6;i++)
{
printf("%s\t%d\t%s\n",subj[i],sub[i],sub[i]>=35?"P":"F");
```

```
}  
  
puts("_____  
_____"");  
  
printf("%s Tot %d Avg %.2f and got ",name,tot,avg);  
  
if(p==0) puts("Failed");  
  
else if(avg>=75)puts("Distinction");  
  
else if(avg>=60)puts("1st Class");  
  
else if(avg>=50)puts("2nd Class");  
  
else puts("3rd Class");  
  
puts("_____  
_____"");  
  
getch();  
  
}
```

```
TC
Enter rollno abc123
Enter stu name peter
Enter Gender Male
Enter Date of birth 10-12-2003
Enter Father name Suresh
Enter Mother name Mary
Enter School name St'anns
Enter Identification mark 1 a more on fore head
Enter Identification mark 2 a mole on fore arm
Enter 6 sub marks 55 66 54 44 45 30

BOARD OF SECONDARY EDUCATION
ANDHRA PRADESH

Rollno: abc123 Name: peter
Gender: Male Date of Birth: 10-12-2003
Father: Suresh Mother: Mary
School: St'anns

Identification Marks

1. a more on fore head
2. a mole on fore arm

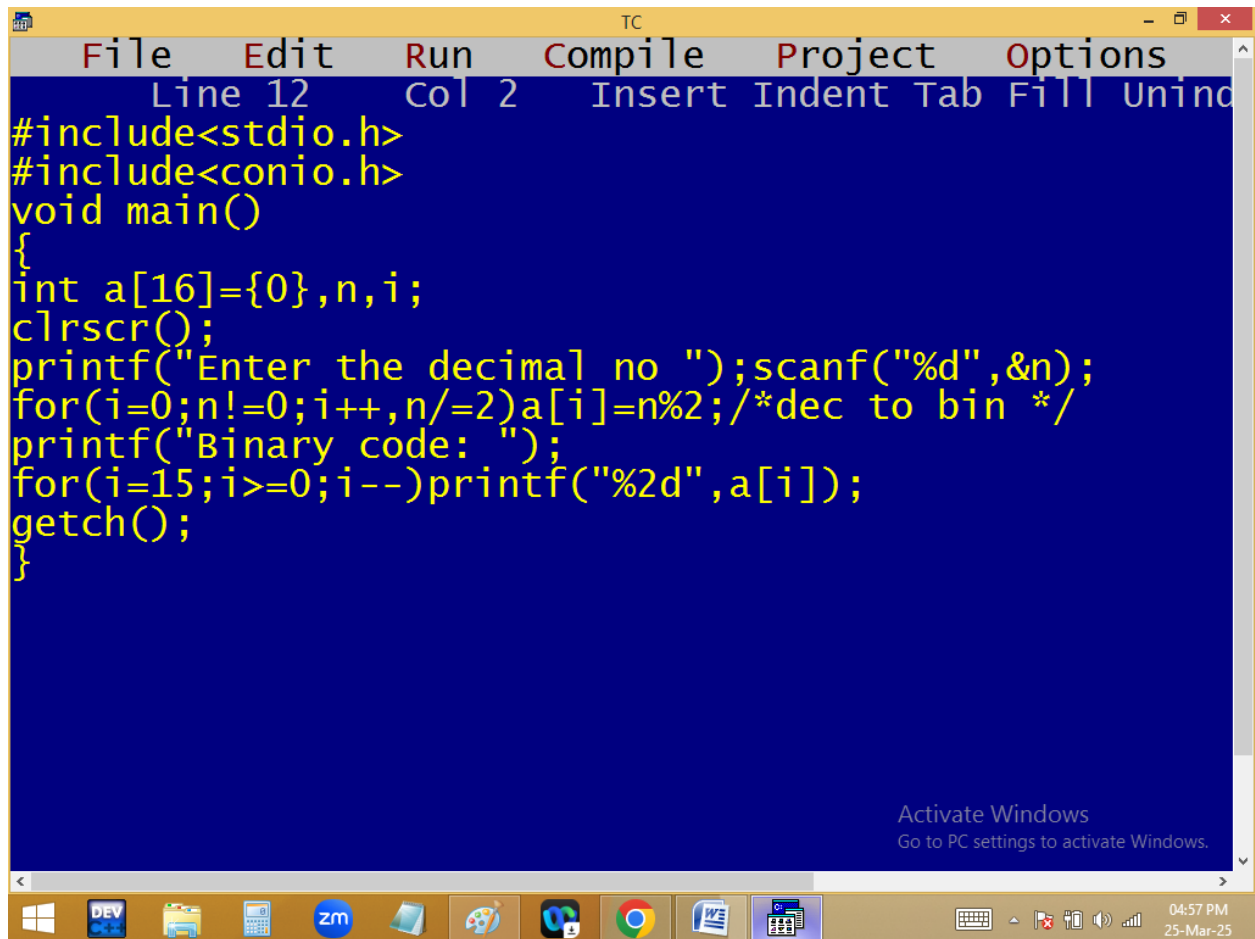
Subject Marks Pass/Fail
Tel 55 P
Eng 66 P
Hin 54 P
Mat 44 P
Sci 45 P
Soc 30 F

peter Tot 294 Avg 49.00 and got Failed

Activate Windows
Go to PC settings to activate Windows.
```

Decimal to binary conversion:

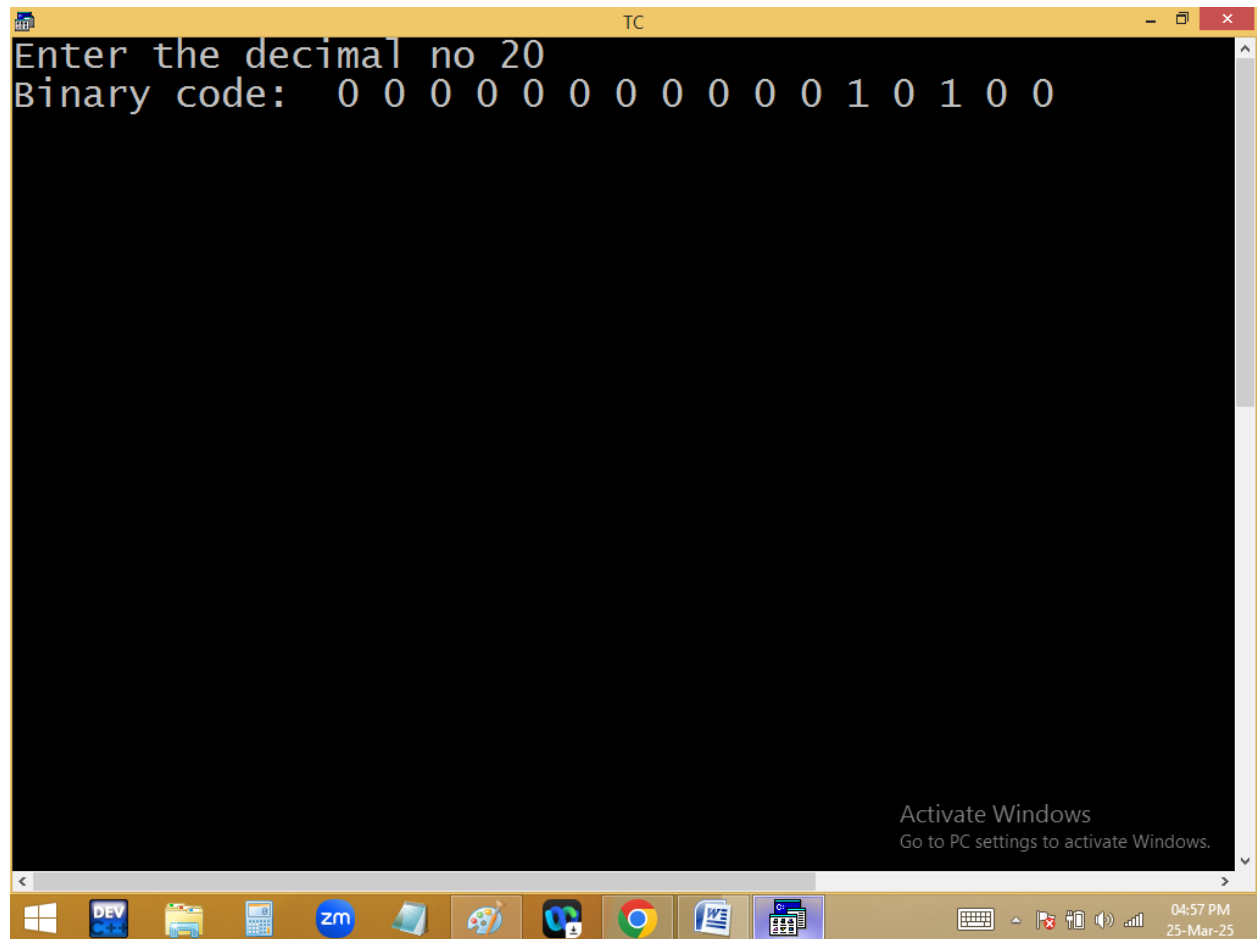
Input 20 → 0000 0000 0001 0100



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, and Options. A status bar at the top of the code area shows "Line 12 Col 2" and some menu options like "Insert", "Indent", "Tab", "Fill", and "Unind". The main area of the window has a dark blue background with yellow text representing C code. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter the decimal no ");scanf("%d",&n);
for(i=0;n!=0;i++,n/=2)a[i]=n%2;/*dec to bin */
printf("Binary code: ");
for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

At the bottom of the window, there is a Windows taskbar with various icons including Windows, DEV, File Explorer, Calculator, Zoom, and others. On the right side of the taskbar, the system clock shows "04:57 PM" and the date "25-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

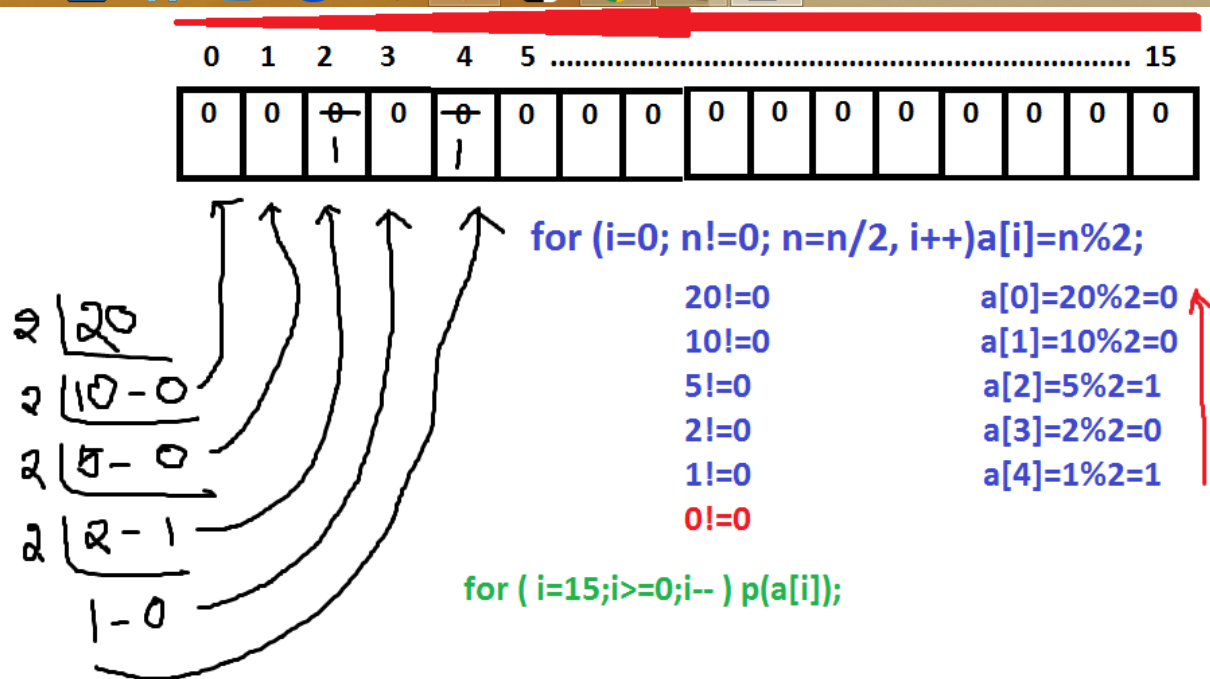


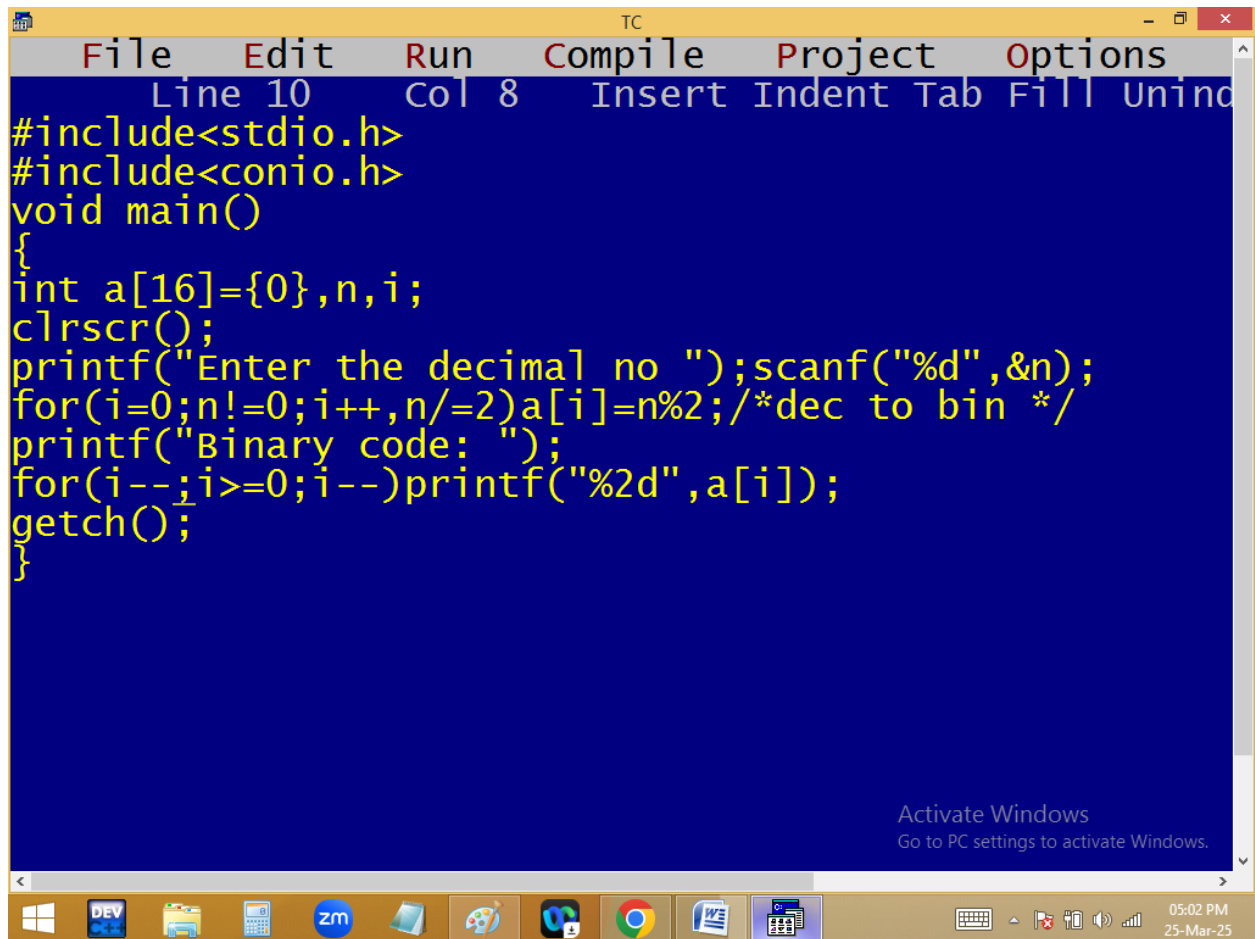
Enter the decimal no 2

Binary code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0

Activate Windows
Go to PC settings to activate Windows.

04:57 PM
25-Mar-25

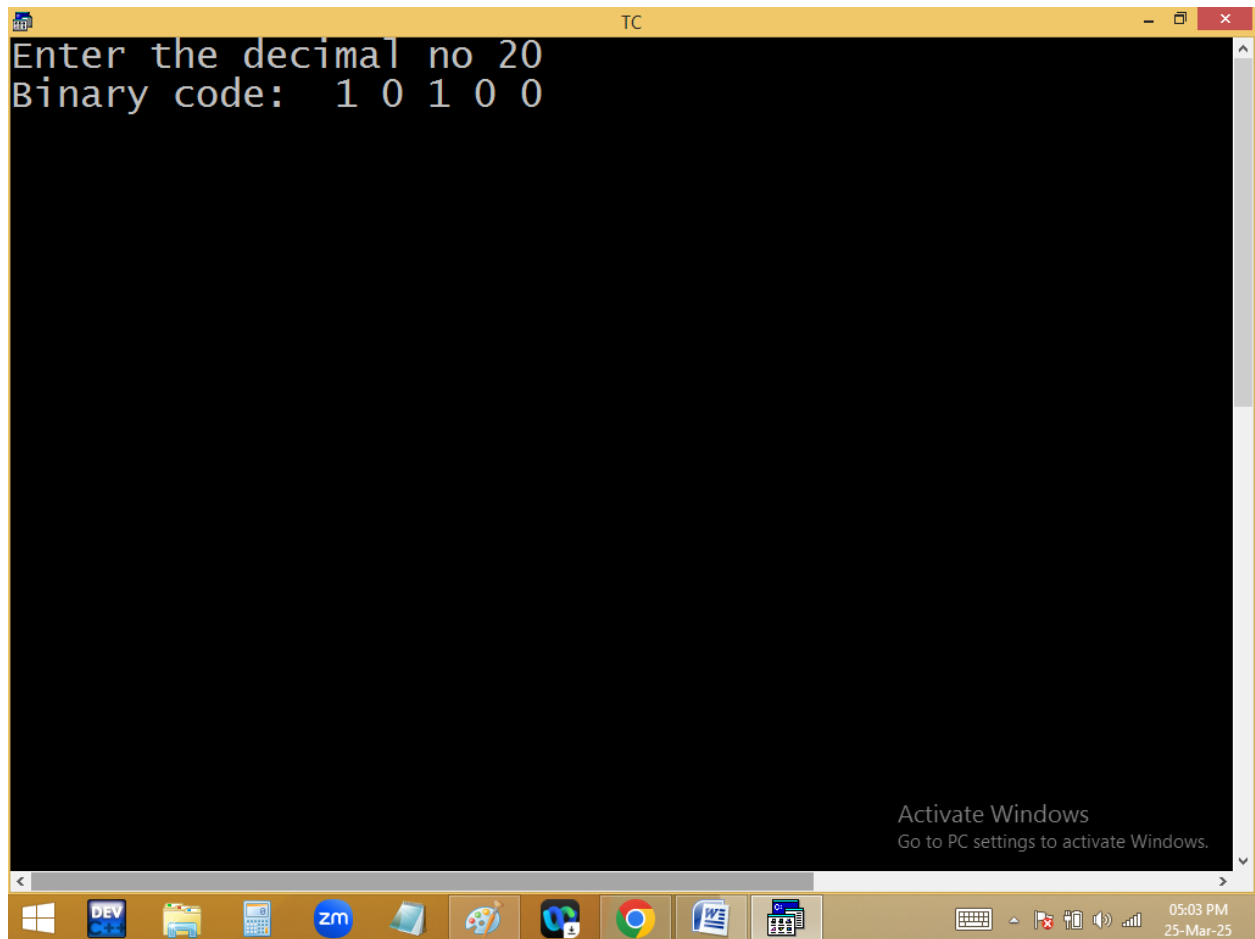




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". Below the menu bar, the status bar shows "Line 10", "Col 8", and "Insert Indent Tab Fill Unind". The main editing area has a blue background with yellow text containing the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter the decimal no ");scanf("%d",&n);
for(i=0;n!=0;i++,n/=2)a[i]=n%2;/*dec to bin */
printf("Binary code: ");
for(i--;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

At the bottom right of the IDE window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time: "05:02 PM 25-Mar-25".

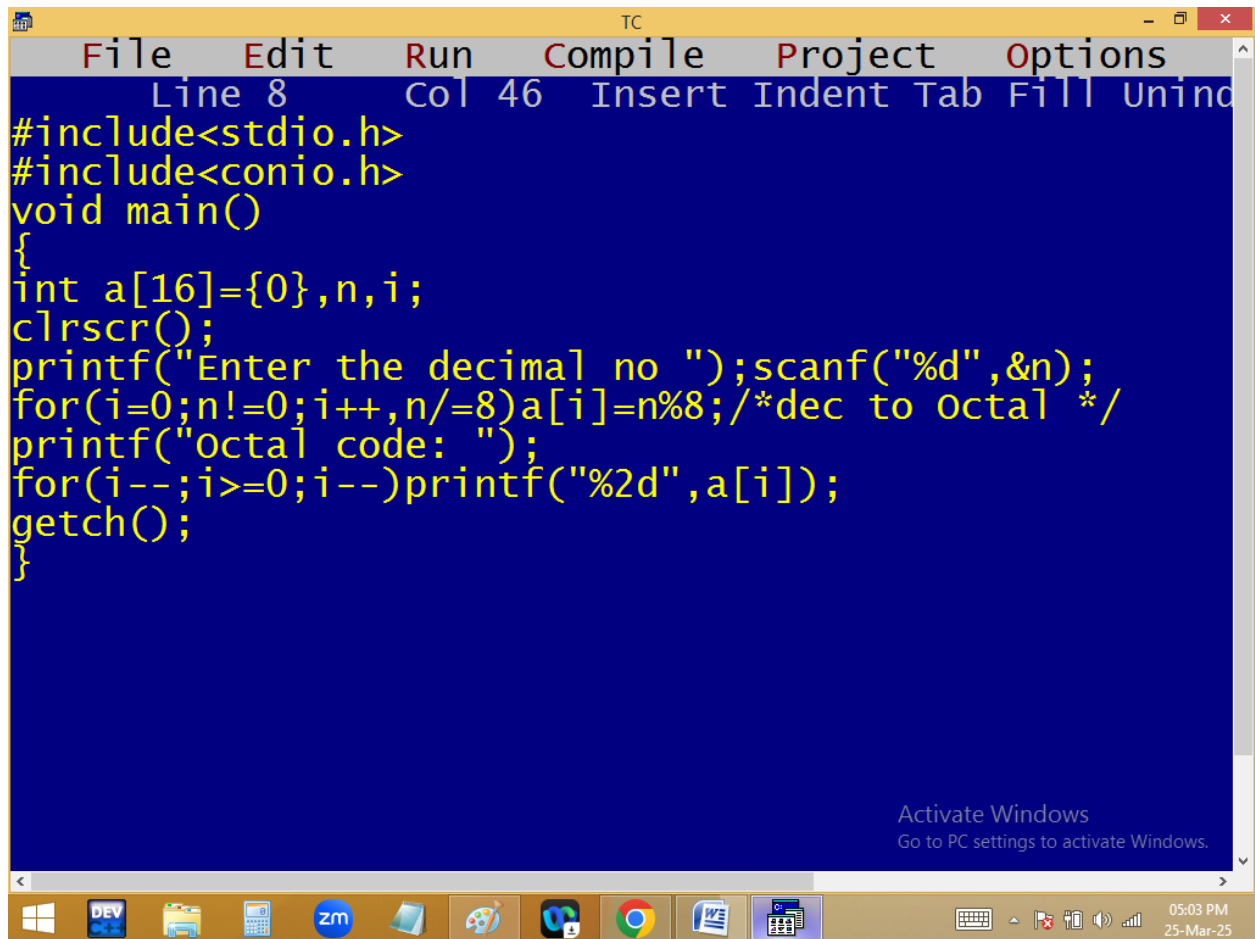


```
TC
Enter the decimal no 20
Binary code: 1 0 1 0 0
```

Activate Windows
Go to PC settings to activate Windows.

05:03 PM
25-Mar-25

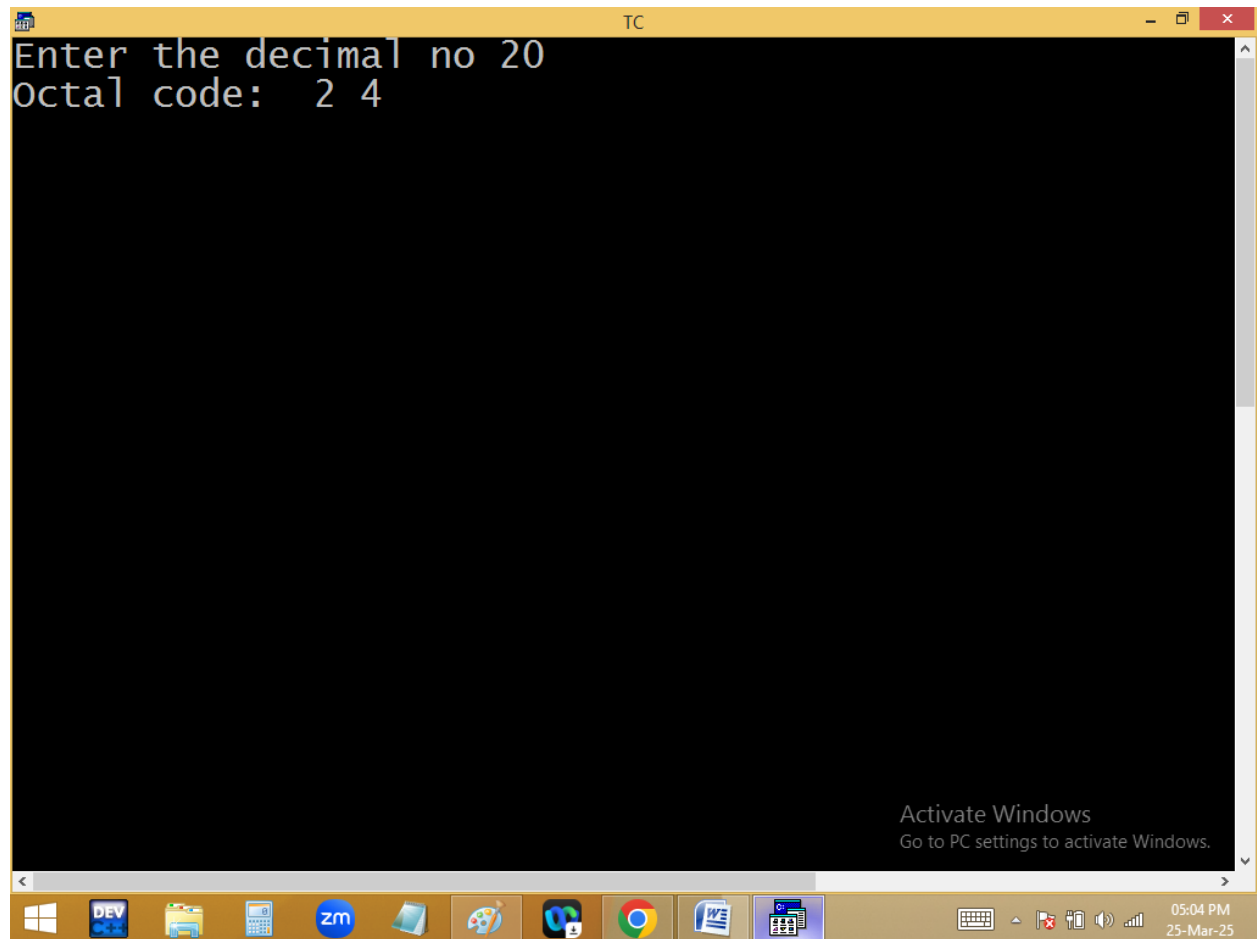
Decimal to octal:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". Below the menu bar, the status bar shows "Line 8" and "Col 46". The main editing area has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter the decimal no ");scanf("%d",&n);
for(i=0;n!=0;i++,n/=8)a[i]=n%8;/*dec to Octal */
printf("Octal code: ");
for(i--;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

At the bottom right of the IDE window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and several other applications. The system clock in the bottom right corner shows "05:03 PM 25-Mar-25".



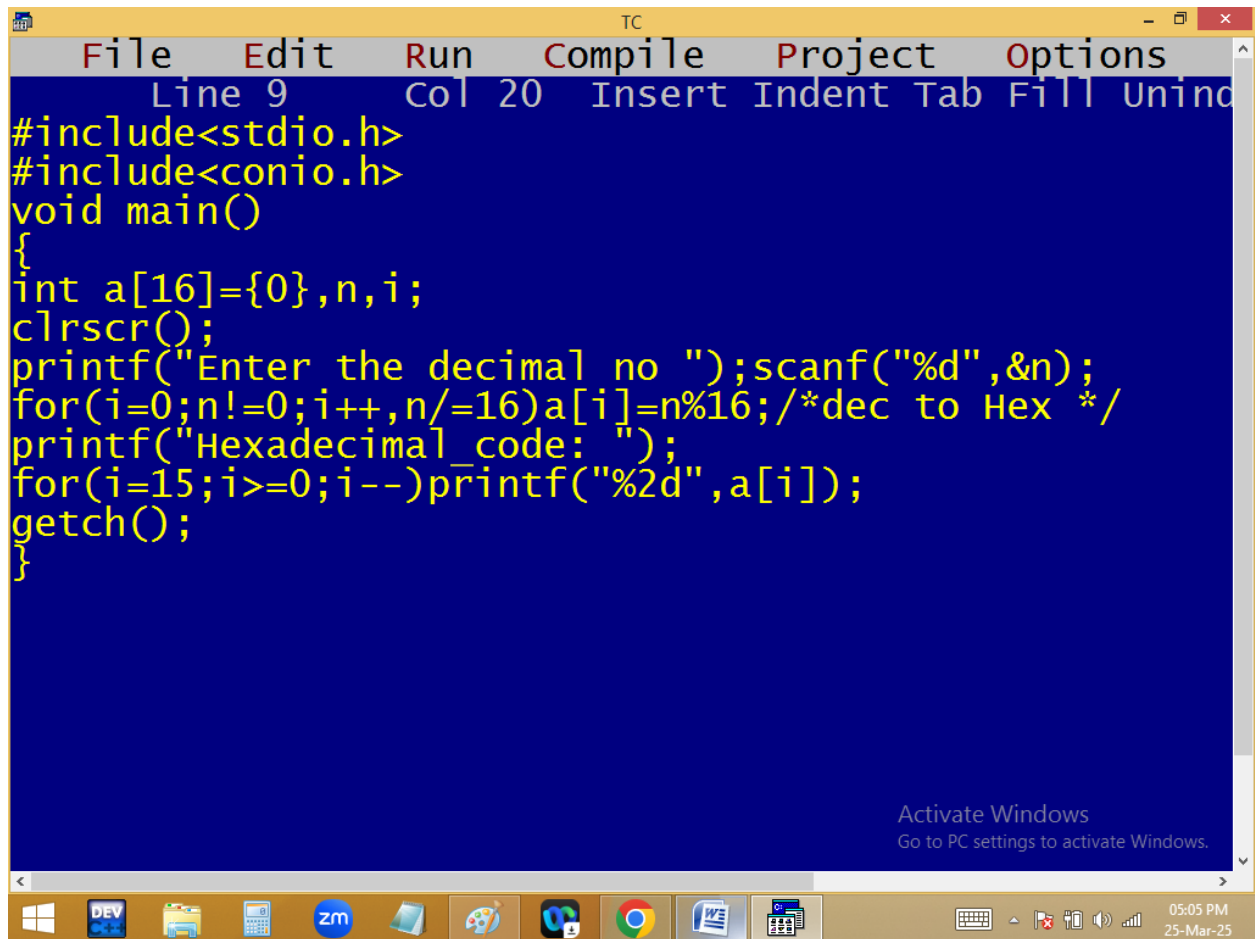
```
TC
Enter the decimal no 50
Octal code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 6 2_

Activate Windows
Go to PC settings to activate Windows.
```

$$\begin{array}{r} 8 \overline{) 20} \\ 2 - 4 \checkmark \end{array}$$

Decimal to hexadecimal:

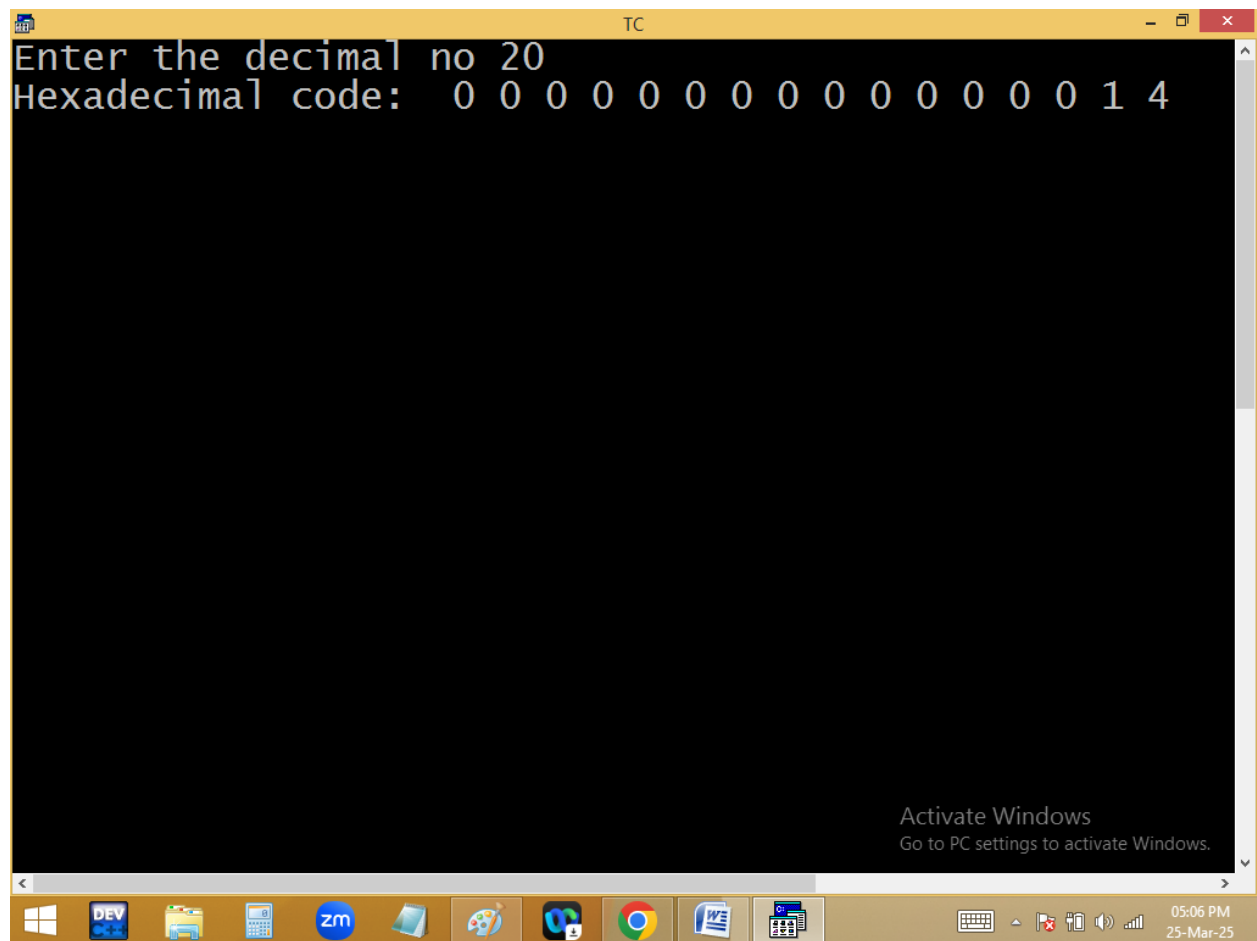
$$16 \overline{) 20} \\ 1-4 \checkmark$$

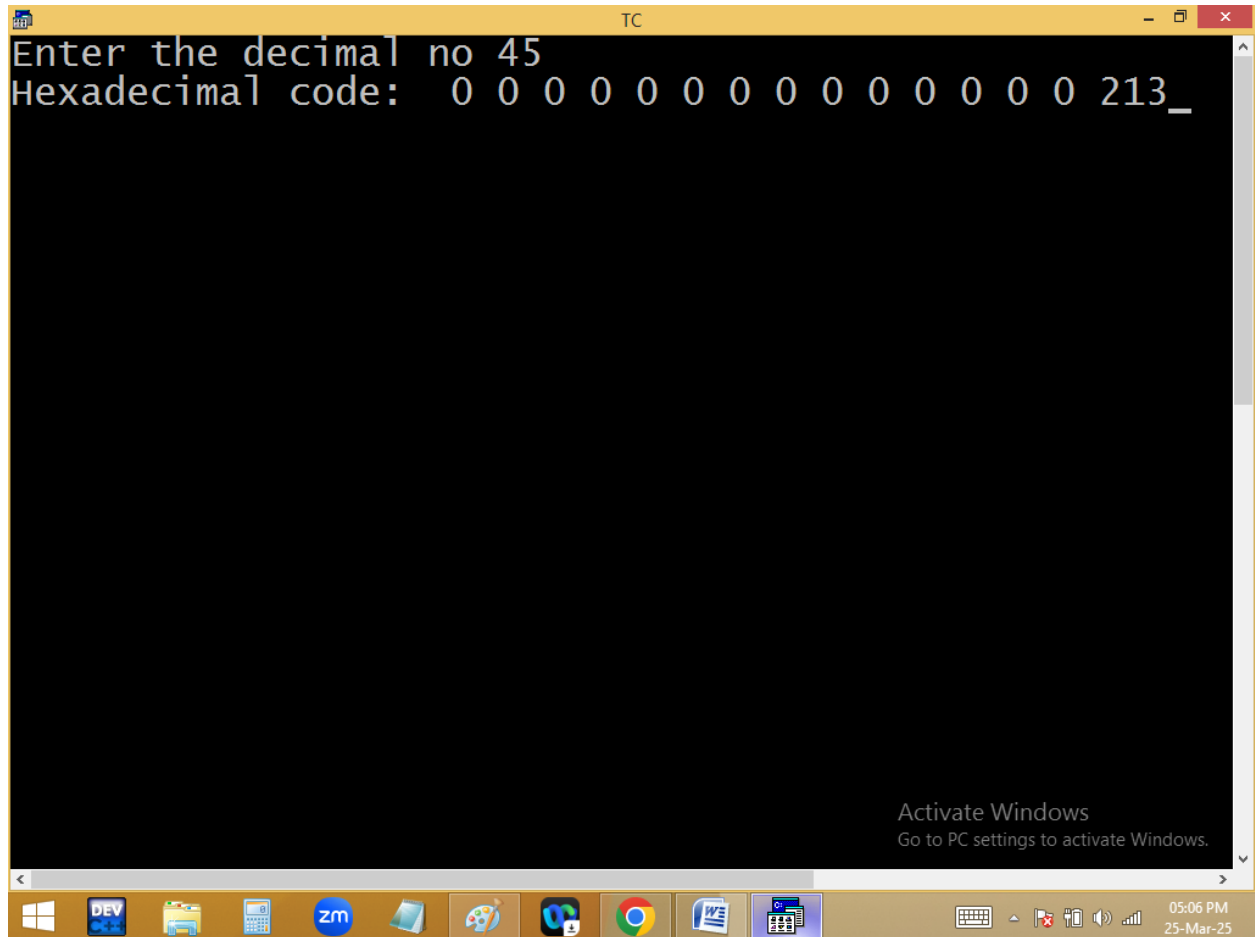


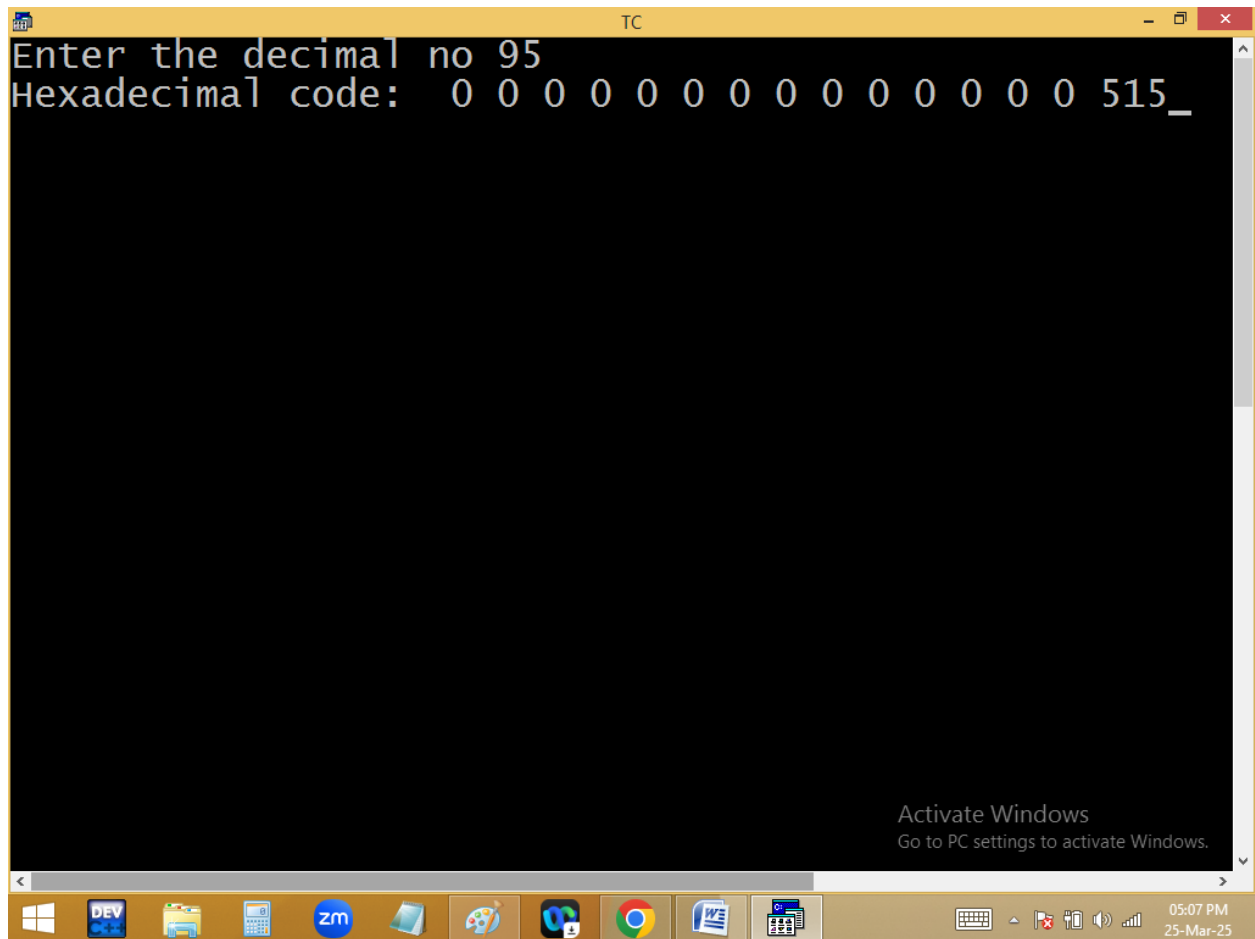
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". Below the menu bar, the status bar shows "Line 9", "Col 20", and "Insert Indent Tab Fill Unind". The main editing area has a dark blue background with yellow text. The code is as follows:

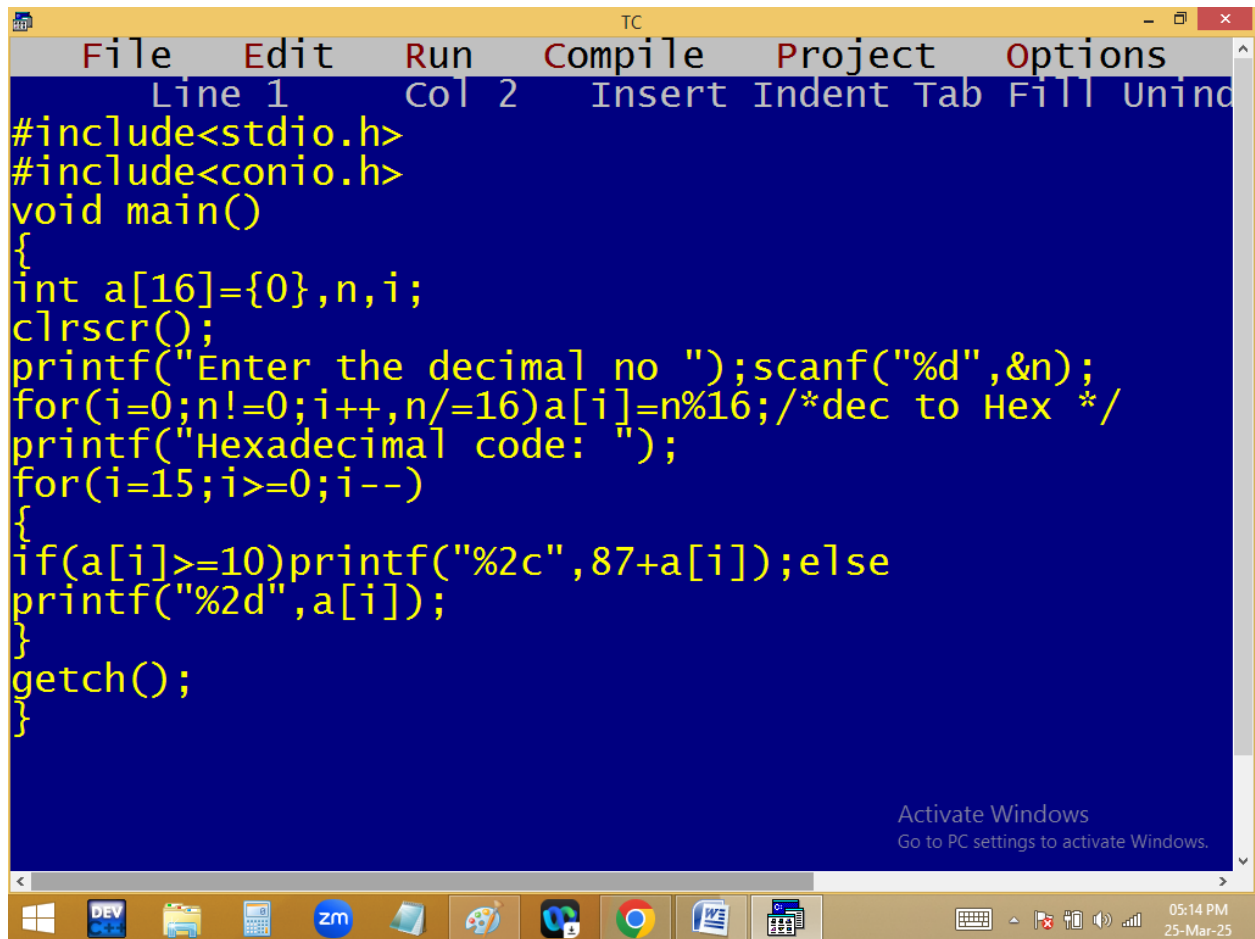
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter the decimal no ");scanf("%d",&n);
for(i=0;n!=0;i++,n/=16)a[i]=n%16;/*dec to Hex */
printf("Hexadecimal_code: ");
for(i=15;i>=0;i--)printf("%2d",a[i]);
getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, a folder, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time as "05:05 PM 25-Mar-25".





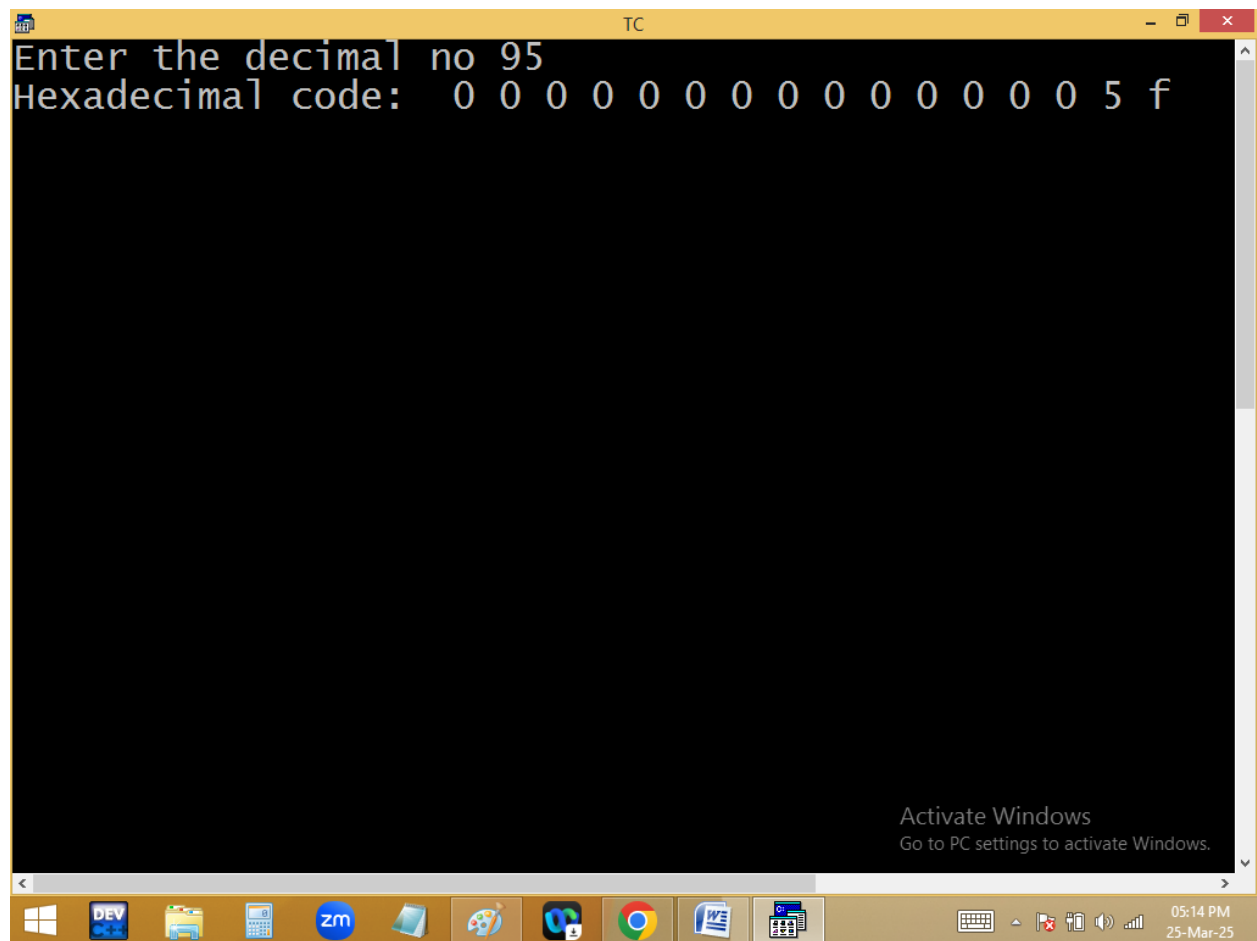


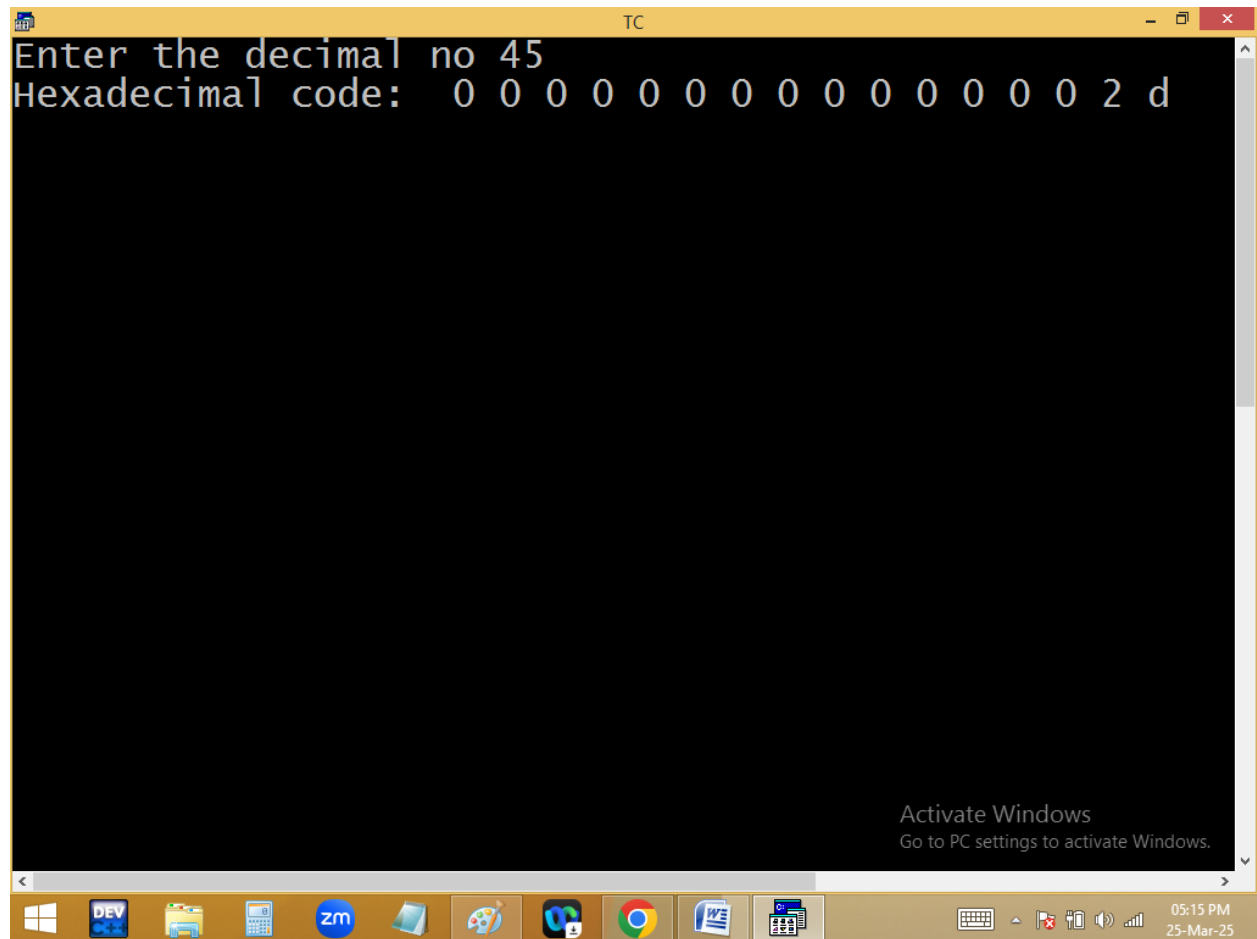


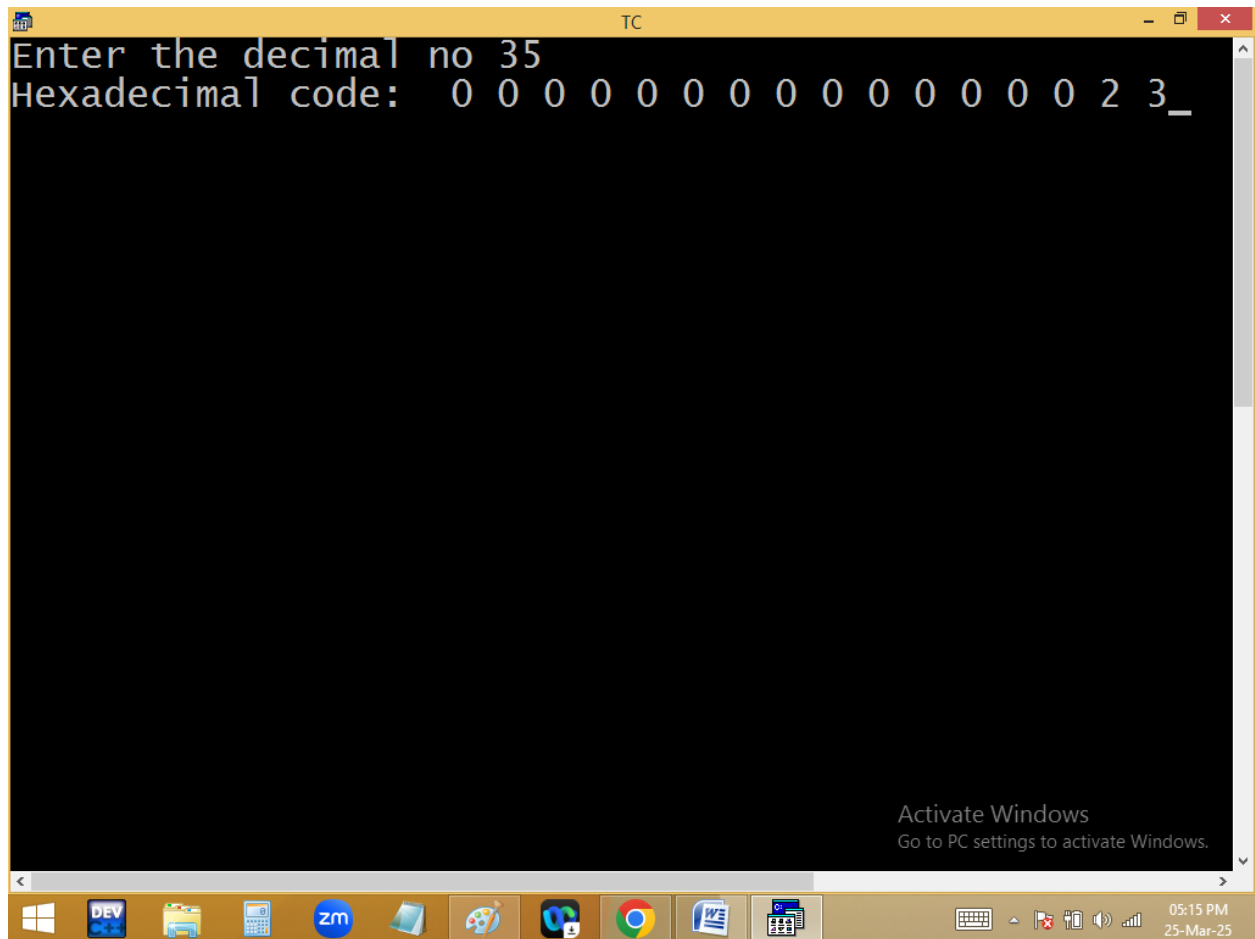
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". Below the menu bar, the status bar shows "Line 1" and "Col 2". The main editing area has a dark blue background with yellow text. The code is a C program that converts a decimal number to its hexadecimal representation. The code is as follows:

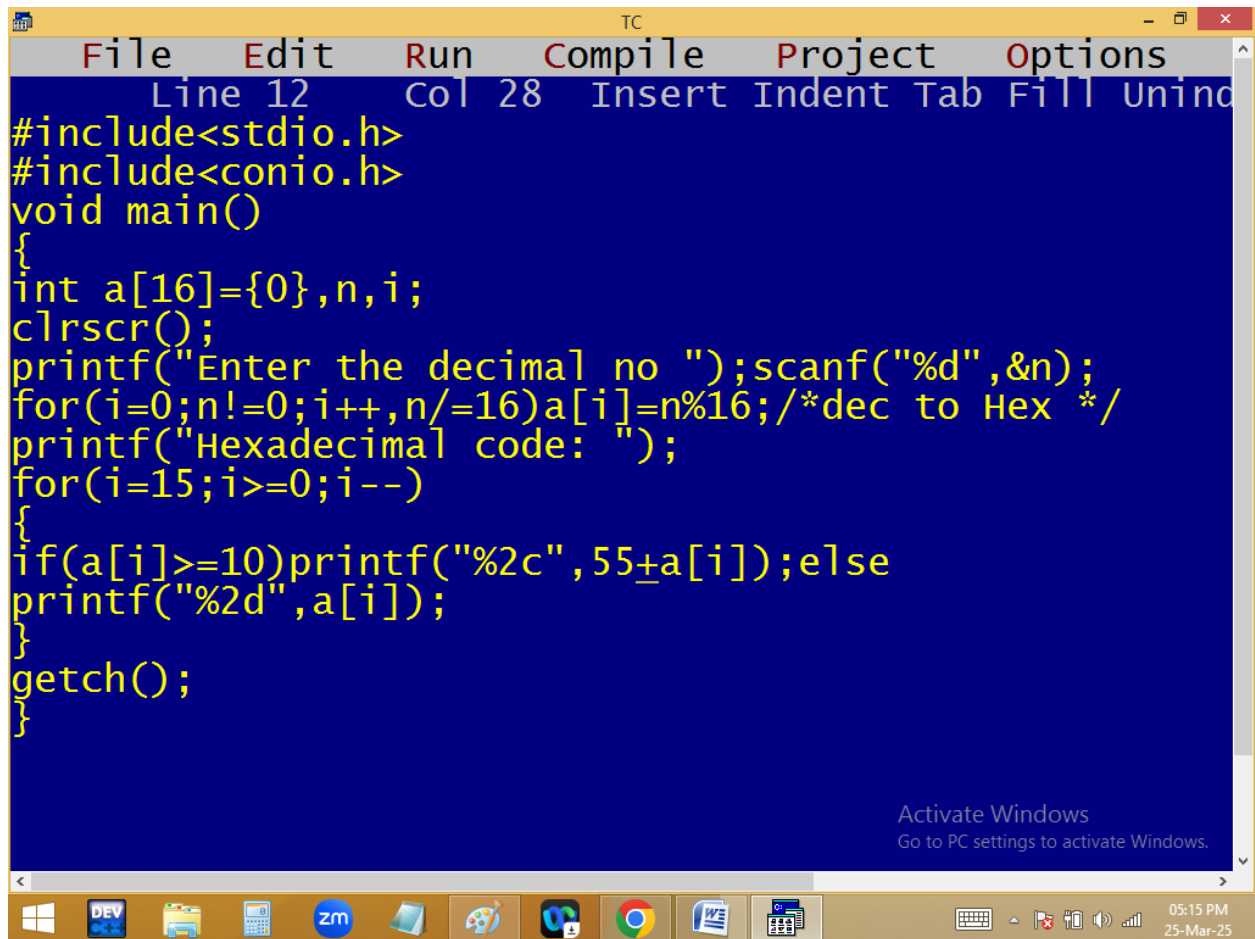
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter the decimal no ");scanf("%d",&n);
for(i=0;n!=0;i++,n/=16)a[i]=n%16;/*dec to Hex */
printf("Hexadecimal code: ");
for(i=15;i>=0;i--)
{
if(a[i]>=10)printf("%2c",87+a[i]);else
printf("%2d",a[i]);
}
getch();
}
```

At the bottom right of the IDE window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 05:14 PM on 25-Mar-25.









The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". The status bar at the top indicates "Line 12", "Col 28", and "Insert Indent Tab Fill Unind". The main editing area has a blue background and contains the following C code:

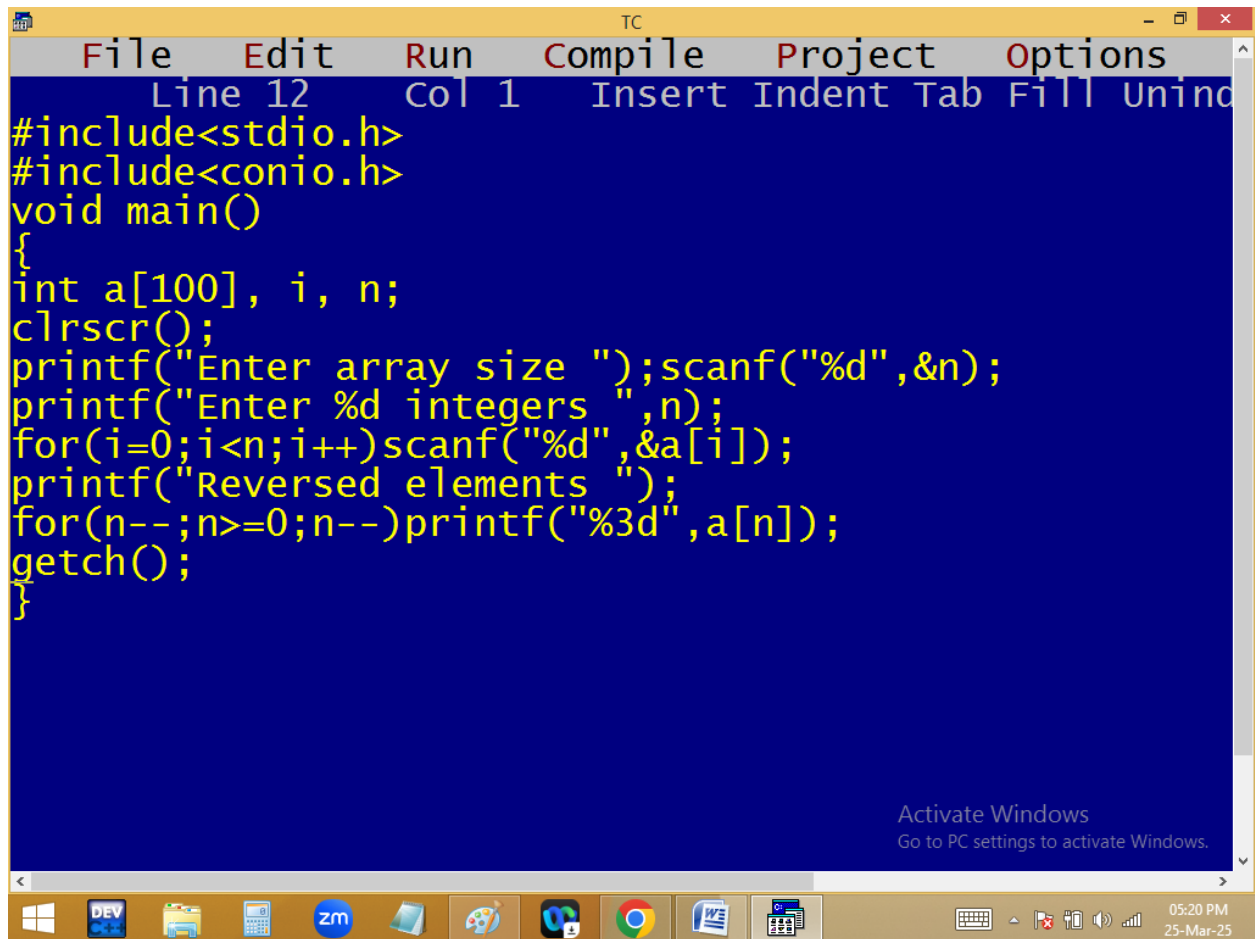
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[16]={0},n,i;
clrscr();
printf("Enter the decimal no ");scanf("%d",&n);
for(i=0;n!=0;i++,n/=16)a[i]=n%16;/*dec to Hex */
printf("Hexadecimal code: ");
for(i=15;i>=0;i--)
{
if(a[i]>=10)printf("%2c",55+a[i]);else
printf("%2d",a[i]);
}
getch();
}
```

An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, stating "Go to PC settings to activate Windows." The Windows taskbar is at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the time as 05:15 PM on 25-Mar-25.

The image shows a screenshot of a Turbo C++ (TC) compiler window. The title bar reads "TC". The main text area contains the following text:
Enter the decimal no 95
Hexadecimal code: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 5 F
At the bottom right of the window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and several other applications. The system clock in the bottom right corner shows "05:16 PM 25-Mar-25".

Arranging array elements in reverse order:

Temporary:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". Below the menu bar, the status bar shows "Line 12", "Col 1", and "Insert Indent Tab Fill Unind". The main editing area has a blue background with yellow text containing the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100], i, n;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Reversed elements ");
for(n--;n>=0;n--)printf("%3d",a[n]);
getch();
}
```

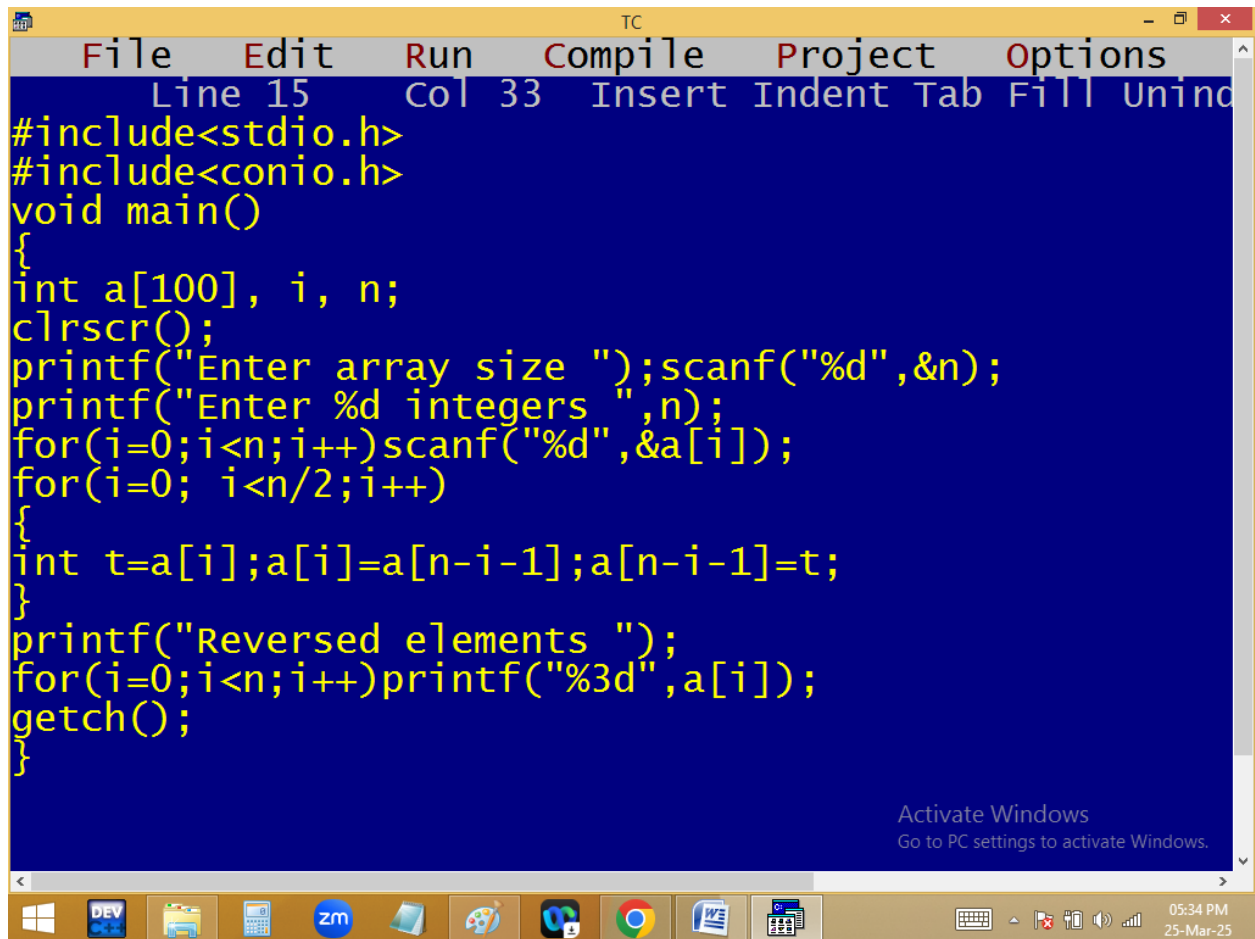
At the bottom right of the window, there is a watermark that says "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the very bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and several other applications. The system clock in the bottom right corner indicates "05:20 PM 25-Mar-25".

```
TC
Enter array size 5
Enter 5 integers 1 2 3 4 5
Reversed elements 5 4 3 2 1_
```

Activate Windows
Go to PC settings to activate Windows.

05:20 PM
25-Mar-25

Permanent arrangement:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", and "Options". Below the menu bar, a status bar shows "Line 15", "Col 33", and "Insert Indent Tab Fill Undo". The main editing area has a blue background with yellow text. The code is a C program that reverses an array. It includes `<stdio.h>` and `<conio.h>`, defines `main()`, declares an array `a` of size 100 and variables `i` and `n`, calls `clrscr()`, prompts the user for the array size `n` and then `n` integers, and finally prints the reversed elements. The task is completed by swapping elements from the start and end of the array towards the center. The Windows taskbar at the bottom shows various application icons and the system clock indicating 05:34 PM on 25-Mar-25.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100], i, n;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<n/2;i++)
{
int t=a[i];a[i]=a[n-i-1];a[n-i-1]=t;
}
printf("Reversed elements ");
for(i=0;i<n;i++)printf("%3d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

```
TC
Enter array size 5
Enter 5 integers 9 5 1 7 2
Reversed elements 2 7 1 5 9_
```

Activate Windows
Go to PC settings to activate Windows.

05:34 PM
25-Mar-25

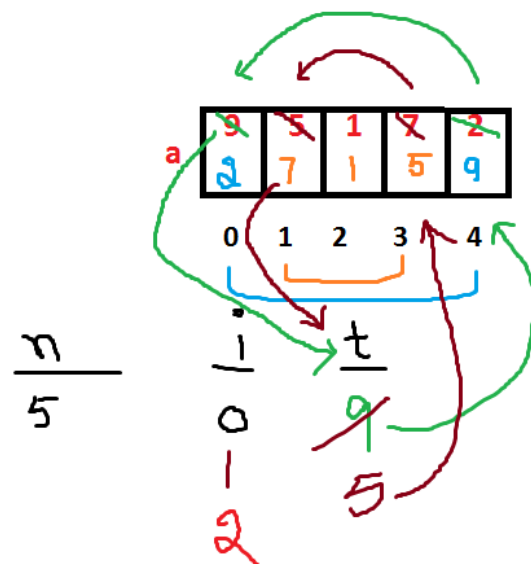
```
TC
Enter array size 4
Enter 4 integers 1 0 2 4
Reversed elements 4 2 0 1_

Activate Windows
Go to PC settings to activate Windows.
```

```
for( i=0; i<n/2; i++)
{
t = a[0];
a[0] = a[5-0-1];

a[5-0-1]=t;
}
```

```
t = a[1];
a[1]=a[5-1-1]
a[5-1-1]=t;
```



Without using temp variable [no 3rd variable]:

```
TC
File Edit Run Compile Project Options
Line 13 Col 20 Insert Indent Tab Fill Undo
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100], i, n;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
for(i=0; i<n/2;i++)
{
a[i]=a[i]+a[n-i-1];a[n-i-1]=a[i]-a[n-i-1];
a[i]=a[i]-a[n-i-1];
}
printf("Reversed elements ");
for(i=0;i<n;i++)printf("%3d",a[i]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

```
TC
Enter array size 6
Enter 6 integers 1 2 3 4 5 6
Reversed elements 6 5 4 3 2 1_
```

Activate Windows
Go to PC settings to activate Windows.

05:42 PM
25-Mar-25


```

Enter array size 3
Enter 3 integers 1 4 3
Reversed elements 3 4 1

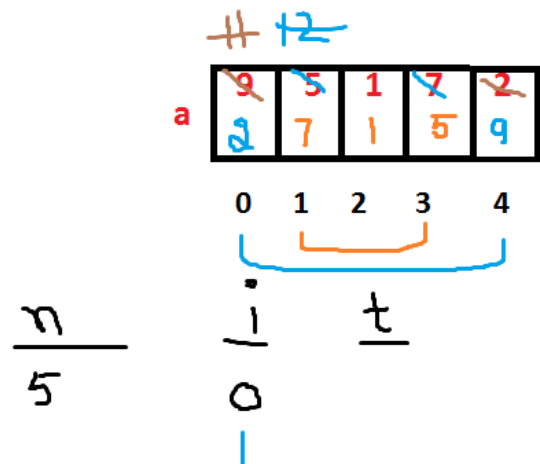
```

Activate Windows
Go to PC settings to activate Windows.

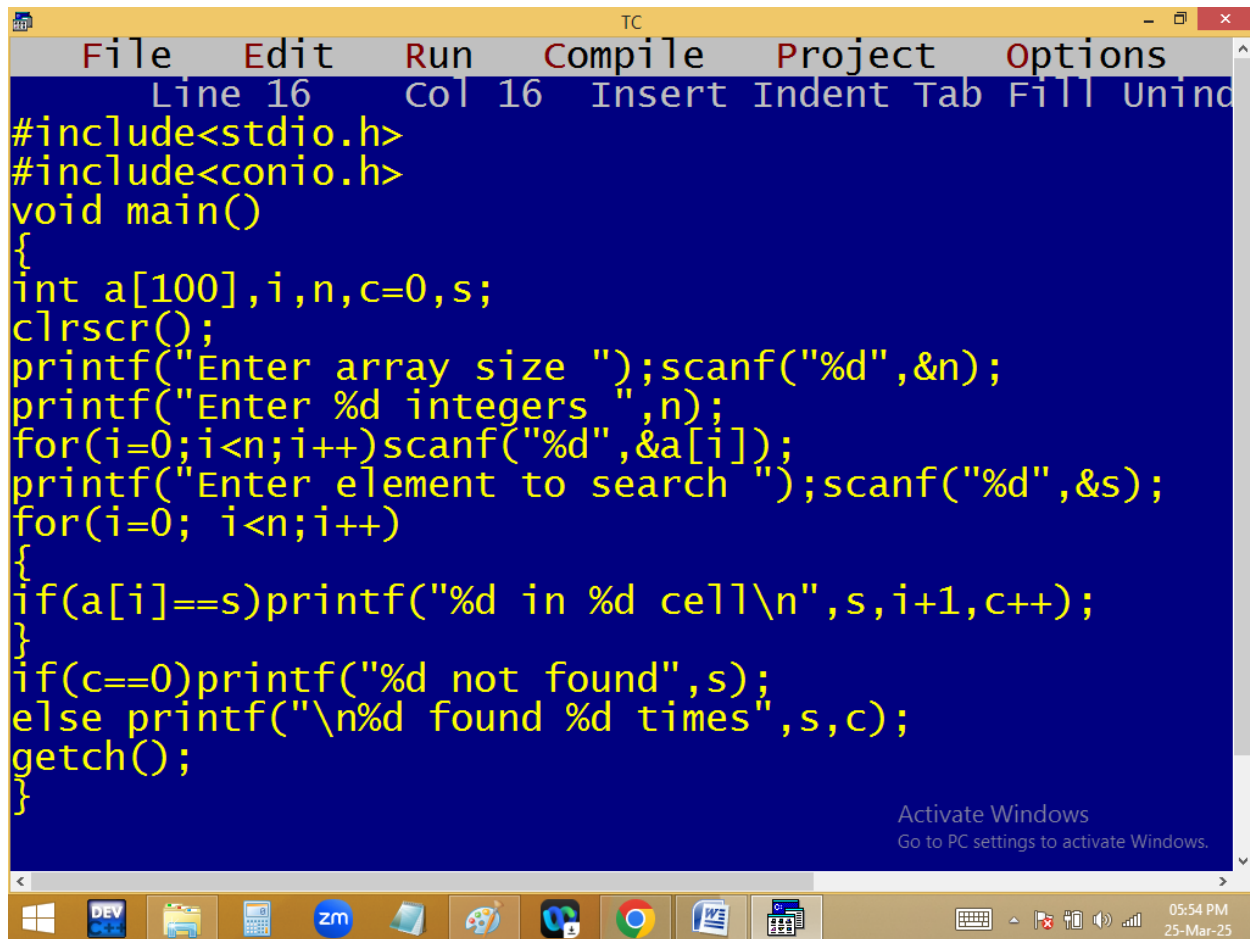
```

for( i=0; i<n/2; i++ )
{
a[0]=9+2=11
a[i]=a[i]+a[n-i-1];
a[1]=a[1] + a[3]==>5+7=12
a[4]=a[0]-a[4]==>11-2=9
a[n-i-1]=a[i]-a[n-i-1];
a[3]=a[1]-a[3]==>12-7=5
a[0]=a[0]-a[4]==>11-9=2
a[i]=a[i]-a[n-i-1];
a[1]=a[1]-a[3]==>12 - 5 = 7
}

```



Linear search:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a menu bar with 'File', 'Edit', 'Run', 'Compile', 'Project', and 'Options'. Below the menu bar, it displays 'Line 16 Col 16' and some menu options like 'Insert', 'Indent', 'Tab', 'Fill', and 'Unind'. The main editing area has a dark blue background with yellow text. The code is a C program that takes an array of integers and a search element, then prints the frequency of the search element in the array. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[100],i,n,c=0,s;
clrscr();
printf("Enter array size ");scanf("%d",&n);
printf("Enter %d integers ",n);
for(i=0;i<n;i++)scanf("%d",&a[i]);
printf("Enter element to search ");scanf("%d",&s);
for(i=0; i<n;i++)
{
if(a[i]==s)printf("%d in %d cell\n",s,i+1,c++);
}
if(c==0)printf("%d not found",s);
else printf("\n%d found %d times",s,c);
getch();
}
```

At the bottom right of the IDE window, there is a message: 'Activate Windows Go to PC settings to activate Windows.' The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock indicating 05:54 PM on 25-Mar-25.

```
TC
Enter array size 9
Enter 9 integers 1 0 -3 7 1 4 9 1 -2
Enter element to search 1
1 in 1 cell
1 in 5 cell
1 in 8 cell

1 found 3 times
```

Activate Windows
Go to PC settings to activate Windows.

05:54 PM
25-Mar-25

```
TC
Enter array size 3
Enter 3 integers 1 2 3
Enter element to search 0
0 not found_
```

Activate Windows
Go to PC settings to activate Windows.

05:54 PM
25-Mar-25

